

Comparative Analysis of BERT and BART for Text Summarization: Choosing the Optimal Model for Abstractive Summarization

CB.AI.U4AID23026
(Navya SA)

CB.AI.U4AID23040
(Sahithi Dommaraju)

CB.AI.U4AID23046
(Thulasi Jyothi Reddy)

Abstract

With the exponential growth of digital content, the need for efficient text summarization techniques has become increasingly critical. Automated text summarization has emerged as an essential tool for reducing the cognitive load on users by condensing large documents into concise, informative summaries. Text summarization, a fundamental task in Natural Language Processing (NLP), involves condensing a given text into a shorter version while retaining its essential information. Two prominent transformer-based models, BERT (Bidirectional Encoder Representations from Transformers) and BART (Bidirectional and Auto-Regressive Transformers), have demonstrated significant capabilities in this field. BERT, primarily designed for bidirectional contextual understanding, is well-suited for extractive summarization but lacks the ability to generate novel text. In contrast, BART, a denoising autoencoder with an autoregressive decoder, excels in abstractive summarization by generating fluent and coherent summaries. This report provides a comprehensive analysis of BERT and BART, comparing their architectures, training methodologies, and summarization performance. The choice of BART for this project is justified based on its superior ability to produce human-like, concise, and contextually relevant summaries. By leveraging BART's abstractive summarization capabilities, this project aims to enhance the readability and accessibility of large volumes of text, ensuring efficient information retrieval and comprehension.

Introduction

The rapid digitalization of information has led to an unprecedented increase in the volume of textual data. From research papers and news articles to legal documents and customer reviews, extracting relevant insights from massive textual sources has become a major challenge. Automated text summarization has emerged as an essential tool for reducing the cognitive load on users by condensing large documents into concise, informative summaries.

Text summarization can be broadly categorized into extractive and abstractive approaches. Extractive summarization involves selecting key sentences or phrases directly from the source text, preserving their original wording and structure. On the other hand, abstractive summarization generates new sentences that paraphrase the main ideas of the document, providing more coherent and natural-sounding summaries. While extractive methods were traditionally preferred due to their simplicity and accuracy, the advent of deep learning models has enabled significant advancements in abstractive summarization, producing more fluent and contextually rich summaries.

Among the state-of-the-art transformer-based models in NLP, BERT (Bidirectional Encoder Representations from Transformers) and BART (Bidirectional and Auto-Regressive Transformers) have gained widespread recognition for their ability to process and summarize text effectively. BERT, introduced by Google in 2018, revolutionized NLP with its bidirectional understanding of text,

improving contextual awareness in language tasks. However, its primary application in extractive summarization limits its effectiveness for tasks requiring text generation. In contrast, BART, developed by Facebook AI, builds upon the strengths of BERT while incorporating an autoregressive decoder, making it highly suitable for abstractive summarization.

This report aims to explore the working mechanisms of BERT and BART, highlighting their differences in summarization capabilities. It will also justify the selection of BART for this project based on its superior performance in abstractive summarization. By implementing BART, the objective is to produce more readable, contextually relevant, and fluent summaries, thereby improving the accessibility and usability of large textual datasets.

Extractive vs. Abstractive Summarization

Text summarization is categorized into two main approaches: extractive and abstractive summarization. Extractive summarization works by identifying and selecting key sentences or phrases directly from the original text, preserving their structure and wording. This method relies on ranking sentences based on importance, often using statistical, machine learning, or deep learning models like BERT. Extractive techniques ensure high factual accuracy since they do not alter the original text but may lack coherence and fluency.

On the other hand, abstractive summarization generates new sentences that paraphrase the essential information from the input text. Instead of merely extracting key phrases, it understands the meaning and restructures content in a human-like manner. Abstractive summarization, enabled by advanced NLP models like BART and T5, often produces more natural, concise, and coherent summaries. However, due to its generative nature, it poses challenges like factual consistency and requires more sophisticated language understanding.

BERT: *Bidirectional Encoder Representations from Transformers*

BERT, introduced by Google in 2018, is a transformer-based model designed to understand the context of words in a sentence bidirectionally. Unlike traditional left-to-right or right-to-left language models, BERT considers both preceding and succeeding words in a sentence, enhancing its ability to capture complex linguistic dependencies. This bidirectional approach enables BERT to excel in numerous NLP tasks, including question answering, named entity recognition, sentiment analysis, and extractive summarization.

BERT Architecture

BERT's architecture is based on the transformer encoder structure, comprising multiple layers of self-attention mechanisms and feedforward neural networks. The model is pre-trained using two primary techniques:

1. **Masked Language Model (MLM):**
During pre-training, a portion of the input tokens is randomly masked, and BERT learns to predict these missing words based on their surrounding context.
2. **Next Sentence Prediction (NSP):**
BERT is trained to determine whether two given sentences appear sequentially in a document. This helps the model grasp contextual relationships between consecutive sentences.

BERT is available in different variations, including BERT-Base (12 encoder layers, 110 million parameters) and BERT-Large (24 encoder layers, 340 million parameters). These variations provide trade-offs between computational efficiency and performance.

Limitations of BERT for Summarization

While BERT performs well in extractive summarization, it lacks the ability to generate new words or restructure sentences in a meaningful way. Extractive summarization often fails to create fluent and coherent summaries, making it unsuitable for tasks that require a more natural and human-like summarization approach. This limitation paved the way for the development of models better suited for text generation, such as BART.

BART: Bidirectional and Auto-Regressive Transformers

BART, introduced by Facebook AI, is a transformer-based model specifically designed for text generation tasks, including summarization. It combines the strengths of both BERT and GPT (Generative Pre-trained Transformer), making it a hybrid model capable of bidirectional context understanding and autoregressive text generation.

BART Architecture

BART follows a denoising autoencoder architecture, consisting of two main components:

1. **Encoder:** Similar to BERT, BART's encoder is bidirectional, allowing it to capture contextual relationships effectively.
2. **Decoder:** Unlike BERT, BART includes an autoregressive decoder, similar to GPT, which generates text token by token.

BART is trained using a novel approach in which the input text is intentionally corrupted by applying various noise functions, such as:

- **Token deletion:** Removing random words from the input sequence.
- **Token masking:** Hiding some tokens to encourage contextual learning.
- **Sentence shuffling:** Reordering sentences to make the model learn structural coherence.

- **Text infilling:** Replacing spans of text with a single mask token.

By training the model to reconstruct the original text from its corrupted version, BART develops strong capabilities in text generation, making it highly effective for abstractive summarization.

Advantages of BART for Summarization

1. **Superior Abstractive Summarization:** BART generates concise and coherent summaries rather than simply extracting key sentences from the input text.
2. **Robust Pretraining Strategy:** The denoising autoencoder mechanism allows BART to handle noisy, incomplete, or unstructured text more effectively.
3. **Fluency and Coherence:** BART's autoregressive decoder produces natural-sounding summaries that closely resemble human-written text.
4. **Performance on Benchmarks:** BART consistently outperforms BERT-based models on summarization benchmarks such as CNN/Daily Mail and XSum.
5. **Versatility:** Beyond summarization, BART is highly effective for other NLP tasks, including machine translation, question answering, and text completion.

Why We Choose BART for Text Summarization?

Given the strengths of BART in abstractive summarization, it was chosen over BERT for this project due to the following reasons:

- **Ability to Generate Novel Text:** BART can rephrase content rather than merely extracting sentences.

- **Higher Readability:** The model produces well-structured and coherent summaries with improved fluency.
- **Better Handling of Complex Texts:** BART is more adept at summarizing documents that require contextual paraphrasing and restructuring.
- **Improved Performance in NLP Tasks:** Studies show that BART consistently outperforms BERT in text summarization benchmarks, making it the more suitable model for this application.

Conclusion

The field of text summarization has seen significant advancements with the introduction of deep learning models, particularly transformer-based architectures. BERT and BART, two of the most influential models, have transformed NLP applications with their robust text-processing capabilities. While BERT excels in extractive summarization by leveraging bidirectional contextual understanding, it lacks the ability to generate novel text, limiting its applicability in abstractive summarization. On the other hand, BART's denoising autoencoder architecture and autoregressive decoding enable it to produce coherent, fluent, and human-like summaries, making it the ideal choice for summarization tasks that require paraphrasing and sentence restructuring.

The decision to use BART for this project is grounded in its superior performance in abstractive summarization. Its ability to handle

noisy and unstructured text, generate concise and meaningful summaries, and outperform BERT-based models on various NLP benchmarks makes it the optimal solution. By leveraging BART, this project aims to enhance information accessibility, improve knowledge retention, and facilitate efficient text comprehension. As NLP continues to evolve, further advancements in transformer models and summarization techniques will contribute to even more accurate, context-aware, and high-quality text generation, reinforcing the significance of automated summarization in an increasingly data-driven world.

REFERENCES:

- [1] Attada Venkataramana, K. Srividya, and R. Cristin, "Abstractive Text Summarization Using BART," in 2022 IEEE 2nd Mysore Sub Section International Conference (MysuruCon), 2022.
- [2] Lewis, M., Liu, Y., Goyal, N., Ghazvininejad, M., Mohamed, A., Levy, O., Stoyanov, V., & Zettlemoyer, L. (2019). BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation, Translation, and Comprehension. *Proceedings of ACL 2020*, 7871–7880.
- [3] Liu, Y., & Lapata, M. (2019). Text Summarization with Pretrained Encoders. *Transactions of the Association for Computational Linguistics*, 7, 705–719.